



Further Function

Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

1 A circle on the number plane has equation $x^2 + y^2 - 8x + 6y + 21 = 0$.

a. Find the centre and radius of the circle. 2

b. Give the domain of the circle.

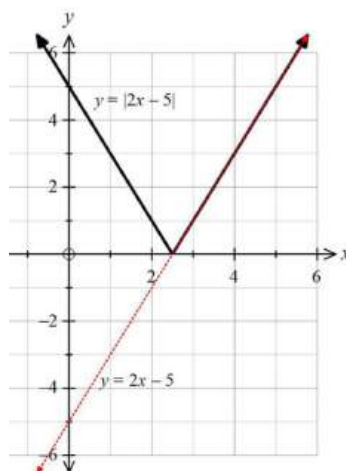
a) Recognise the curve as a circle, so complete the square:

$$\begin{aligned} x^2 + y^2 - 8x + 6y + 21 &= 0 \\ x^2 - 8x + y^2 + 6y &= -21 \\ x^2 - 8x + 4^2 + y^2 + 6y + 3^2 &= 16 + 9 - 21 \\ x^2 - 8x + 16 + y^2 + 6y + 9 &= 4 \\ (x - 4)^2 + (y + 3)^2 &= 2^2 \end{aligned}$$

Curve is a circle, centre $(4, -3)$ and radius 2

b) From Sketch $2 \leq x \leq 6$

2 Draw a neat sketch of $y = |2x - 5|$. 1



3 Two functions are defined as follows: 2

$$h(x) = x^3 \text{ and } j(x) = \frac{2x-1}{3}$$

Find the value of $h(j(5))$.

$$j(x) = \frac{2x-1}{3}$$

OR

$$j(5) = \frac{2(5)-1}{3}$$

$$= 3$$

$$\begin{aligned} h(3) &= 3^3 \\ &= 27 \end{aligned}$$

$$h(j(x)) = \left(\frac{2x-1}{3} \right)^3$$

$$\begin{aligned} h(j(5)) &= \left(\frac{2(5)-1}{3} \right)^3 \\ &= (3)^3 \\ &= 27 \end{aligned}$$

4 The diagram below shows the graph of a cubic function. What is the equation of this cubic function? 1

A. $y = -x(x+1)(x-2)$

B. $y = -x(x-1)(x+2)$

C. $y = -x(x+1)(x-2)$

D. $y = -x(x-1)(x+2)$

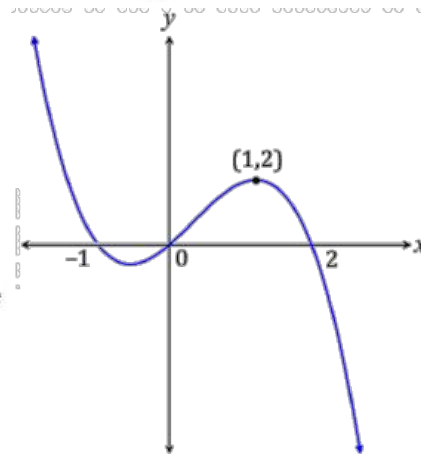
Intercepts are -1, 0 and 2.

$$y = -x(x+1)(x-2)$$

Test point (1, 2)

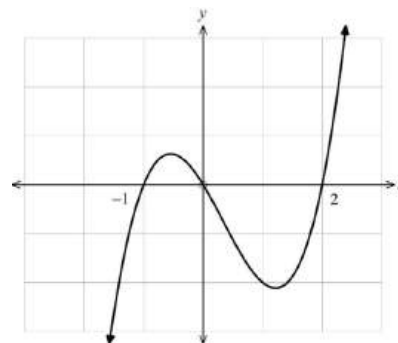
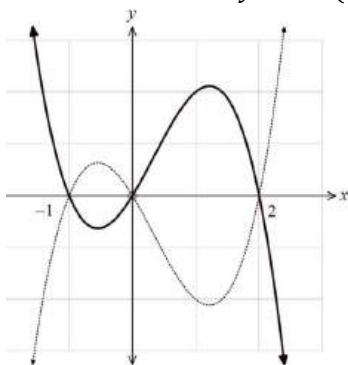
$$2 = -1 \times (1+1) \times (1-2) \text{ True}$$

A



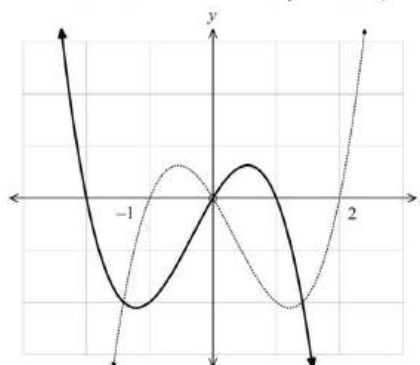
5 The sketch of $y = k(x)$ is shown on the right.

a. Draw a neat sketch of $y = -k(x)$.



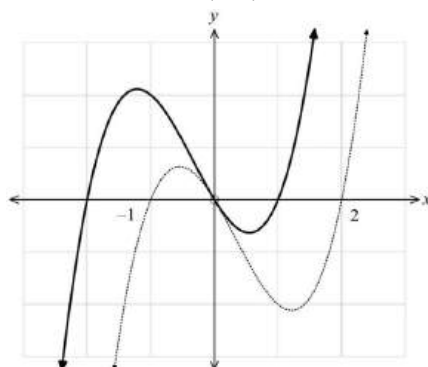
1

b. Draw a neat sketch of $y = k(-x)$.



1

c. Draw a neat sketch of $y = -k(-x)$.



1

6 If $f(x) = \sqrt{4-x}$ and $g(x) = x^2$,

a. Find the composite function $f(g(x))$.

1

b. Find the domain and range of the composite function.

2

a. $f(g(x)) = \sqrt{4-x^2}$

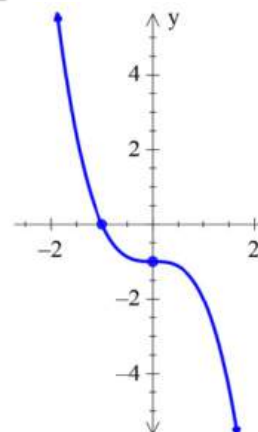
b.

Domain : $x^2 \leq 4 \quad \therefore -2 \leq x \leq 2$, written as $[-2, 2]$ in interval notation

Range : $0 \leq y \leq 2$, written as $[0, 2]$ in interval notation

7 Given $f(x) = x^3 - 1$ sketch the graph of $f(-x)$.

2



8 Find the values of x where the graph of the function $f(x) = \frac{1}{x^2-4x}$ is discontinuous.

2

$$f(x) = \frac{1}{x(x-4)} \quad \therefore \text{discontinuities at } x=0 \text{ and } x=4$$

9 Solve $|x + 5| = 5$.

$$x = -10 \quad \text{or} \quad x = 0$$

2

10 For the function $y = x^2 + 2x - 3$.

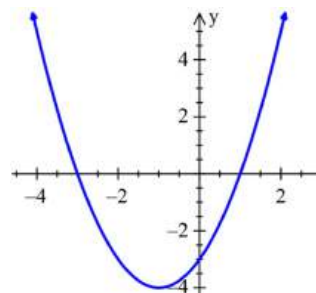
- a. Calculate the discriminant and use it to determine the number of roots of the function.

1

=16 **Positive discriminant: 2 real roots**

- b. Sketch the function, showing all important features

(Vertex, x-intercepts, y-intercepts).



2

11 It is given that $1 < x < 4$. What is the value of $|x - 1| - |x - 4|$?

1

$$1 < x \Rightarrow x - 1 > 0 \Rightarrow |x - 1| = x - 1$$

$$x < 4 \Rightarrow x - 4 < 0 \Rightarrow |x - 4| = 4 - x$$

$$|x - 1| - |x - 4| = x - 1 - 4 + x = 2x - 5$$

12 a. Find the radius and centre of the circle $x^2 - 4x + y^2 + 2y = 0$.

1

b. Find the coordinates of points where $x^2 - 4x + y^2 + 2y = 0$ and $y = 2x - 8$ intersect.

2

a $x^2 - 4x + (4 - 4) + y^2 + 2y + (1 - 1) = 0$

$$x^2 - 4x + 4 + y^2 + 2y + 1 = 5$$

$$(x - 2)^2 + (y + 1)^2 = 5$$

Radius: $\sqrt{5}$, Centre: $(2, -1)$

b Substitute $y = 2x - 8$ in $x^2 - 4x + y^2 + 2y = 0$

$$x^2 - 4x + (2x - 8)^2 + 2(2x - 8) = 0$$

$$5x^2 - 32x + 48 = 0 \Rightarrow x = \frac{12}{5} \text{ or } 4$$

Intersection points: $\left(\frac{12}{5}, -\frac{16}{5}\right)$ and $(4, 0)$

13 The function f is defined by $f(x) = 2 + \sqrt{x - 3}$ for $x \geq 3$. The function g is defined by

$$g(x) = \frac{12}{x} + 2 \text{ for } x > 0.$$

- a. Write the domain and range of the function f using interval notation.

2

domain: $x \geq 3$ $[3, \infty)$
range: $y \geq 2$ $[2, \infty)$

- b. Write an expression for the composite function $h(x) = g(f(x))$ and hence find a value

2

for $g(f(12))$.

$$g(f(x)) = \frac{12}{2 + \sqrt{x - 3}} + 2$$

$$g(f(12)) = \frac{12}{2 + \sqrt{12 - 3}} + 2 = \frac{22}{5}$$



Further Function

Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

1 Solve $|3x + 5| = 7$. $3x + 5 = 7$

$3x = 2$

$x = \frac{2}{3}$

$3x + 5 = -7$

$3x = -12$

$x = -4$

2

- 2 The graph shows $y = f(x)$, which is an incomplete odd function. Which of the following statements about

$y = f(x)$ is correct?

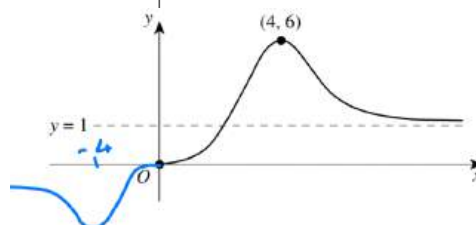
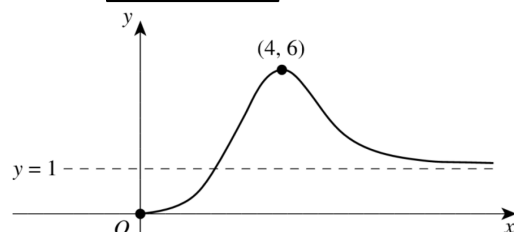
A. $f(x)$ has only one horizontal asymptote at $y = 1$.

B. $f(-4) = 6$

C. $f'(-3) > 0$

D.

D. $f(x)$ is decreasing for $x > 4$ and $-4 < x < 0$.

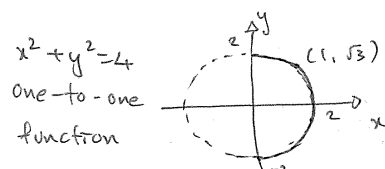


1

- 3 The graph of $f(x) = \log_2 x$ is transformed to produce $g(x) = 1 - \log_2(x - 2)$. Describe, in words, the transformation steps used to obtain the graph of $y = g(x)$.

- a. Reflect in the x-axis
 a. Translate 2 units to the right
 a. Translate 1 unit up

- 4 If $x^2 + y^2 = 4$ is a one-to-one function which includes the point $(1, \sqrt{3})$, what is the largest range of the function?



Range : $-2 \leq y \leq 2$ or $[-2, 2]$
 But this is not in the option.
 So, largest is: $[0, 2]$

1

- 5 The function $f(x)$ is an even function, and the function $g(x)$ is an odd function. Which of the following is a true statement?

A. $g(f(x))$ is neither odd nor even

B. $g(f(x))$ is odd

C. $f(g(x))$ is even

D. $f(g(x))$ is odd

We need to test $g(f(x))$ and $f(g(x))$:

$g(f(-x))$

$= g(f(x))$ (as f is even)

Therefore, $g(f(x))$ is even. (this option is not available as an answer)

$f(g(-x))$

$= f(-g(x))$ (as g is odd)

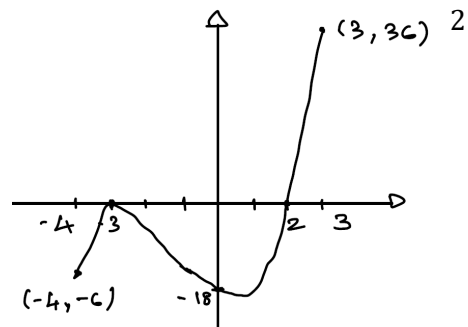
$= f(g(x))$ (as f is even)

Therefore, $f(g(x))$ is even. (this option is available as an answer)

1

- 6 Solve $|2x - 3| \geq 5$. $2x - 3 \leq -5$ and $2x - 3 \geq 5$ (add 3 to both sides) 2
 $2x \leq -2$ and $2x \geq 8$ (divide both sides by 2)
 $x \leq -1$ and $x \geq 4$

- 7 Sketch the function $y = (x - 2)(x + 3)^2$ in the domain $[-4, 3]$.



- 8 The equation of a circle is given as $x^2 + y^2 - 4x + 6y - 3 = 0$.

- a. Find the centre and radius of the circle.

(a) Rearranging and leaving some space for completing the square:
 $x^2 - 4x + y^2 + 6y = 3$ (complete the square for both x and y) 2
 $x^2 - 4x + 4 + y^2 + 6y + 9 = 3 + 4 + 9$ (factorise perfect squares)
 $(x - 2)^2 + (y + 3)^2 = 16$ Thus, the centre is $(2, -3)$, and the radius is 4 units

- b. Find the domain and range of the relation.

(b) Domain: $2 \pm 4 = -2$ to 6 Range: $-3 \pm 4 = -7$ to 1 1
 $[-2, 6]$ $[-7, 1]$
(inequalities for these intervals are also acceptable)

- c. Show that the relation is many-to-many by substituting an appropriate value for x and for y .

(c) When $x = 2$, $(2 - 2)^2 + (y + 3)^2 = 16$ 2
 $(y + 3)^2 = 16$
 $y + 3 = \pm 4$
 $y = -3 \pm 4$
Hence, there are two y -values for one x -value.
When $y = -3$
 $(x - 2)^2 + (3 - 3)^2 = 16$
 $(x - 2)^2 = 16$
 $x - 2 = \pm 4$
 $x = 2 \pm 4$
Hence, there are two x -values for one y -value.

- 9 What is the value of a for which the y - axis is tangent to the circle 1

$$(x - 2)^2 + (y - 5)^2 = a$$

- A. 2 B. 4 C. 5 D. 25

- 10 The diagram below shows the graph of $y = f(x)$. 1

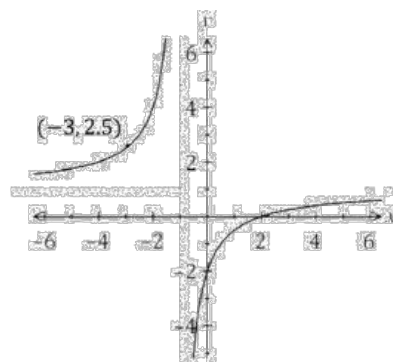
Which of the following statements is false?

- A. The horizontal asymptote is $y = 1$.

- B. The curve is continuous.

- C. The curve is concave up for $x < -1$.

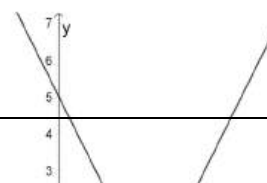
- D. The equation of the function is $y = \frac{x-2}{x+1}$.



- 11 Find the equation of the vertical and horizontal asymptotes of the function $f(x) = \frac{x^2 + 1}{4 - x^2}$. 2

$$x = -2, \quad x = 2, \quad y = -1$$

- 12 Given the graph of $f(x) = |ax + b|$



a. What are the values of a and b ?

2

$$\begin{array}{ll} x \text{ intercept at } 2.5 & y \text{ intercept at } 5 \\ 2.5a - 5 = 0 & |0 + b| = 5 \\ 2.5a = 5 & b = \pm 5 \\ a = 2 & \text{Check a point } (2, 1) \\ & 1 \neq |2 \times 2 + 5| \\ & 1 = |2 \times 2 - 5| \\ & \therefore b = -5 \end{array}$$

$$\therefore a = 2, b = -5$$

b. Solve for $f(x) \geq 3$. $|2x - 5| \geq 3$
 $x \leq 1, x \geq 4$

1

13 Consider the function $y = \frac{8}{x^2 - 4}$.

a. Write down the domain.

Domain: all real x , $x \neq \pm 2$

1

b. Find the horizontal and vertical asymptotes.

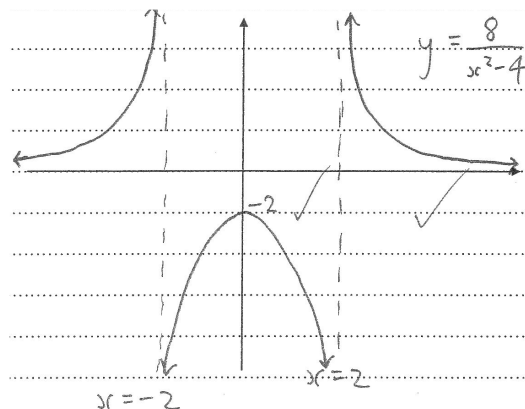
2

Vertical asymptotes: $x = \pm 2$ ✓

Horizontal asymptotes: $\lim_{x \rightarrow \pm \infty} \frac{8}{x^2 - 4} = 0$

$\therefore y = 0$

c. Sketch the function, showing the main features.



2

14 Given the function $f(x) = x^2 - 1$ and $g(x) = \sqrt{4 - x^2}$, sketch $y = f(g(x))$ over its natural domain.

2

$$\begin{aligned} f(g(x)) &= f(\sqrt{4 - x^2}) \\ &= (\sqrt{4 - x^2})^2 - 1 \\ &= 4 - x^2 - 1 \\ &= 3 - x^2 \end{aligned}$$

